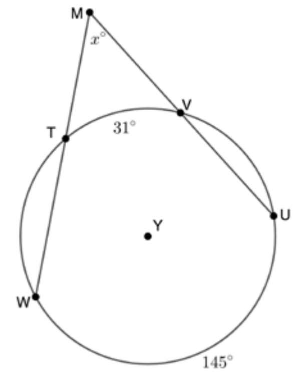
Geometry
Unit 10
Lesson 8 Homework

Name___Answer Key_______
Date____________________
Period___________________

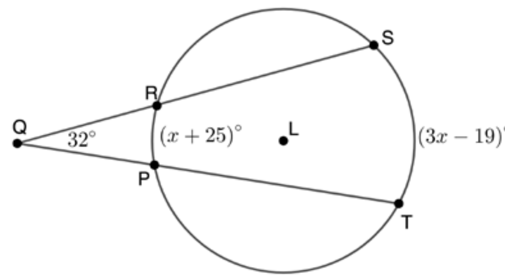
For questions 1-2, Circle Y has points $W, T, V,$ and U on the circle. Secant lines $\overline{WM}$ and $\overline{UM}$ intersect at point M outside the circle. The $m\widehat{UW} = 145°, m\widehat{TV} = 31°$, and $m\angle TMV = x°$.



1. Write a formula that can be used to find the value of x.
   $x = \frac{145-31}{2}$

2. Find the value of x.
   $x = 57$

For questions 3-5, Circle L has points $T, P, R,$ and S on the circle. Secant lines $\overline{TQ}$ and $\overline{SQ}$ intersect at point Q outside the circle. The $m\widehat{ST} = (3x - 19)°$, $m\widehat{PR} = (x + 25)°$, and $m\angle PQR = 32°$.

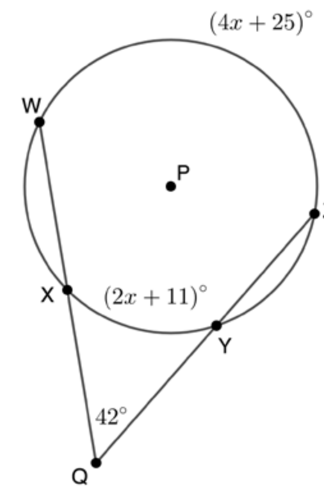


3. Find the value of x.
   $\frac{(3x-19)-(x+25)}{2} = 32$
   $2x - 44 = 64$
   $2x = 108$
   $x = 54$

4. Find the $m\widehat{PR}$.
   $54 + 25 = m\widehat{PR} = 79°$

5. Find the $m\widehat{ST}$.
   $3(54) - 19 = m\widehat{ST} = 143°$

For questions 6-8, Circle P has points $W, X, Y,$ and Z on the circle. Secant lines $\overline{WQ}$ and $\overline{ZQ}$ intersect at point Q outside the circle. The $m\widehat{WZ} = (4x + 25)°$, $m\widehat{XY} = (2x + 11)°$, and $m\angle XQY = 42°$.



6. Find the value of x.

$\frac{(4x+25)-(2x+11)}{2} = 42$

$2x + 14 = 84$

$2x = 70$

$x = 35$

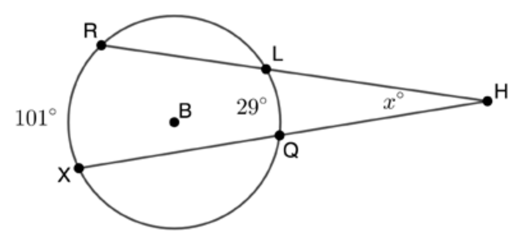
7. Find the $m\widehat{XY}$.

$2(35) + 11 = m\widehat{XY} = 81°$

8. Find the $m\widehat{WZ}$.

$4(35) + 25 = m\widehat{WZ} = 165°$

For questions 9-10, Circle B has points $R, L, Q,$ and X on the circle. Secant lines $\overline{XH}$ and $\overline{RH}$ intersect at point H outside the circle. The $m\widehat{RX} = 101°$, $m\widehat{LQ} = 29°$, and $m\angle LHQ = x°$.



9. Write a formula that can be used to find the value of x.

$x = \frac{101-29}{2}$

10. Find the value of x.

$x = 36$